

2. $\underline{x} \in \mathbb{R}^N$
 $L \subset \mathbb{R}^N$ where basis of L is $\underline{e} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$

$\underline{z} = P_L \underline{x}$ (orthogonal projection of $\underline{x}$ onto L . $\underline{z}$ is the only vector in L s.t. $\underline{x} - \underline{z} \perp L$ i.e. $\langle \underline{x} - \underline{z}, \underline{y} \rangle = 0 \forall \underline{y} \in L$)
 $= \frac{\underline{e}^T \underline{x}}{\underline{e}^T \underline{e}} \underline{e} = \left(\frac{1}{N} \sum_{i=1}^N x_i \right) \underline{e} = [\bar{x}, \dots, \bar{x}] = \bar{\underline{x}}$ i.e. a vector of means of data

$Q_L \underline{x} = \underline{x} - P_L \underline{x}$
 $= \underline{x} - \bar{\underline{x}}$

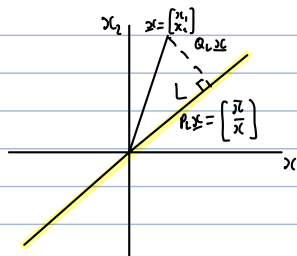
i.e. the difference each obs. is from the mean.

$(\| \underline{y} \|^2 = y_1^2 + y_2^2 + \dots + y_N^2)$

$\Rightarrow \|Q_L \underline{x}\|^2 = \|\underline{x} - \bar{\underline{x}}\|^2$
 $= \langle [x_1 - \bar{x}, x_2 - \bar{x}, \dots, x_N - \bar{x}], [x_1 - \bar{x}, x_2 - \bar{x}, \dots, x_N - \bar{x}] \rangle$
 $= \sum_{i=1}^N (x_i - \bar{x})^2$ i.e. sum of squared deviations

$= (N-1) (\text{Sample Variance})$

$\mathbb{R}^2$:



3. $\| \underline{y} - \underline{x} \|^2 = \| P_M \underline{y} - \underline{x} + Q_M \underline{y} \|^2$
 $= \| P_M \underline{y} - \underline{x} \|^2 + \| Q_M \underline{y} \|^2$
 $= \| P_M \underline{y} - \underline{x} \|^2 + \| \underline{y} - P_M \underline{y} \|^2$
 $\therefore \| \underline{y} - \underline{x} \|^2$ is minimized $\forall \underline{x} \in M$ when $\| \underline{y} - P_M \underline{y} \|^2 = 0$ Since $\| \underline{y} - P_M \underline{y} \|^2 \geq 0$
 $\Rightarrow \inf \{ \| \underline{y} - \underline{x} \|^2 : \underline{x} \in M \} = \| P_M \underline{y} - \underline{x} \|^2 + \| \underline{y} - P_M \underline{y} \|^2$ when $\| \underline{y} - P_M \underline{y} \|^2 = 0$
 $= \| P_M \underline{y} - \underline{x} \|^2$

$\Rightarrow \| \underline{y} - P_M \underline{y} \|^2 = \inf \{ \| \underline{y} - \underline{x} \|^2 : \underline{x} \in M \}$

4. $\underline{y} - \sum_{i=1}^r \frac{\langle \underline{y}, \underline{x}_i \rangle}{\| \underline{x}_i \|^2} \underline{x}_i \perp M$ is equivalent to $\langle \underline{y} - \sum_{i=1}^r \frac{\langle \underline{y}, \underline{x}_i \rangle}{\| \underline{x}_i \|^2} \underline{x}_i, \underline{v} \rangle = 0 \quad \forall \underline{v} \in M$. Since $\underline{v}$ is always a linear combination of the orthogonal basis $\{ \underline{x}_1, \dots, \underline{x}_r \}$, we can instead show $\langle \underline{y} - \sum_{i=1}^r \frac{\langle \underline{y}, \underline{x}_i \rangle}{\| \underline{x}_i \|^2} \underline{x}_i, \underline{x}_j \rangle = 0$ for $j=1, \dots, r$:

$$\langle \underline{y} - \sum_{i=1}^r \frac{\langle \underline{y}, \underline{x}_i \rangle}{\| \underline{x}_i \|^2} \underline{x}_i, \underline{x}_j \rangle = \langle \underline{y}, \underline{x}_j \rangle - \left\langle \sum_{i=1}^r \frac{\langle \underline{y}, \underline{x}_i \rangle}{\| \underline{x}_i \|^2} \underline{x}_i, \underline{x}_j \right\rangle$$

Looking at the terms:

① $\langle \underline{y}, \underline{x}_j \rangle = y_1 x_{1j} + y_2 x_{2j} + \dots + y_n x_{nj}$

② $\left\langle \sum_{i=1}^r \frac{\langle \underline{y}, \underline{x}_i \rangle}{\| \underline{x}_i \|^2} \underline{x}_i, \underline{x}_j \right\rangle = \sum_{i=1}^r \frac{\langle \underline{y}, \underline{x}_i \rangle}{\| \underline{x}_i \|^2} \langle \underline{x}_i, \underline{x}_j \rangle$

$= \frac{\langle \underline{y}, \underline{x}_j \rangle}{\| \underline{x}_j \|^2} \langle \underline{x}_j, \underline{x}_j \rangle$ Since $\langle \underline{x}_i, \underline{x}_j \rangle = 0$ if $j \neq i$ (basis vectors are orthogonal)

$= \frac{\langle \underline{y}, \underline{x}_j \rangle}{\| \underline{x}_j \|^2} \| \underline{x}_j \|^2$

$= \langle \underline{y}, \underline{x}_j \rangle$ for $j=1, \dots, r$

$\Rightarrow \langle \underline{y} - \sum_{i=1}^r \frac{\langle \underline{y}, \underline{x}_i \rangle}{\| \underline{x}_i \|^2} \underline{x}_i, \underline{x}_j \rangle = \langle \underline{y}, \underline{x}_j \rangle - \langle \underline{y}, \underline{x}_j \rangle$

$= 0$ for $j=1, \dots, r$

Hence, $\underline{y} - \sum_{i=1}^r \frac{\langle \underline{y}, \underline{x}_i \rangle}{\| \underline{x}_i \|^2} \underline{x}_i \perp M$